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10.2.3 Deflating the problem

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“ ”

of the diagonal. And our algorithm can then what is called deflate. And we can continue our algorithm by focusing on this submatrix. And we can then pick our shift to always be the bottom right entry of that submatrix. And in that way we will eventually deflate down to a one by one matrix, at which point we have found all of our eigenvalues and eigenvectors. Now because while we are actually converging to this

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Reading assignment

1/1 point (graded)



Read Unit 10.2.3 of the notes. [\[LINK\]](#)

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Homework 10.2.3.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times m}$ be a Hermitian matrix and $V \in \mathbb{C}^{m \times m}$ be a unitary matrix such that

$$V^H A V = \left(\begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right)$$

If V_{00} and V_{11} are unitary matrices such that $V_{00}^H A_{00} V_{00} = \Lambda_0$ and $V_{11}^H A_{11} V_{11} = \Lambda_1$, are both diagonal, show that

$$\left(V \left(\begin{array}{c|c} V_{00} & 0 \\ \hline 0 & V_{11} \end{array} \right) \right)^H A \left(V \left(\begin{array}{c|c} V_{00} & 0 \\ \hline 0 & V_{11} \end{array} \right) \right) = \left(\begin{array}{c|c} \Lambda_0 & 0 \\ \hline 0 & \Lambda_1 \end{array} \right).$$

☒ done



Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Homework 10.2.3.2

1/1 point (graded)
Copy `SimpleShiftedQR.m` into `SimpleShiftedQRAlgWithDeflation` and modify it to add deflation:

```
function [ Ak, V ] = SimpleShiftedQRAlgWithDeflation( A, maxits, illustrate, delay )
```

Modify the appropriate line in [Assignments/Week10/matlab/test_simple_QR_algorithms.m](#), changing `(0)` to `(1)`, and use it to examine the convergence of the method.

☒ done



Solution

For a complete solution, see the [notes](#).

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